

ECON265/CES365

Introduction to Econometrics

Fall 2026
Prof. Alam

Instructor: Moshi Alam

Email: mdalam@clarku.edu

Lectures: Carlson CH128 on Tue & Thu, 10:25–11:40AM

Office Hours: JC222, Thu 12:05–1:05PM, or by appointment

Teaching Assistant (TA): Yujie Zheng

Email: YujiZeng@clarku.edu

Lab/discussion sessions:

Mon 10:00–10:50AM [Carlson CH128]

Wed 10:00–10:50AM [Carlson CH120]

Office Hours: JC201 on Mon & Wed 11:00AM–12:00PM

Exam	Date
Midterm 1	Thursday, October 8, 2026
Midterm 2	Thursday, November 5, 2026
Final exam (cumulative)	Thursday, December 3, 2026

Course description

This course is an Introduction to Econometrics and is a required course for the Economics major. It is also highly recommended for Econ minors and in general, anyone who is interested in using data to understand the world around them better. Materials of ECON 160 (**pre-requisite course**) are expected to be well-understood by students. I strongly recommend students to refresh their concepts of Econ 160 before we deep-dive into Econometrics.

In this course, we will learn concepts and techniques of *econometric methods* in economic analysis and implement them with real data. The empirical models we will build, and use will be guided by intuition of economic models building foundational bridge connecting theory and empirics. Most of the course will focus working with *cross-sectional data*, with a small part towards the end dedicated to *panel data* methods. We will not cover methods related to time-series data in this course. This course will have a strong focus on working with data in R. Throughout the course, in parallel, we will be implementing the concepts learnt using R.

We will begin with a week of reviewing some concepts from ECON160, working with R, followed by introductory concepts of econometrics which initially will build and add mathematical formality to the concepts learned in ECON160, then dive into new econometric methods which deal with *causality*.

I want to emphasize that even though we will be using R heavily, the primary objective in this course is to learn econometric methods, while R is simply a tool to implement them. Hence throughout the course even though I will work with examples with you on R, I will in most cases point you towards resources for you to learn more by yourself since coding is just like learning any language—you will learn the best by

practicing yourself. Thus, attending discussion sections with Yujie—where you will be involved in coding primarily—will be equally important as attending lectures to do well in this course.

Textbook and references

For the econometric methods only, the recommended textbook is the 7th edition of *Introductory Econometrics—A Modern Approach* by Jeffrey Wooldridge. Older versions of the text are acceptable as well but please be aware that chapter numbers may not correspond exactly between different editions. You can also use the copies of the textbook available in the library on reserve. Before delving into the depths of the course materials cover in this course, students must go through **Math Refresher A** (Basic Mathematical Tools) and **B** (Fundamentals of Probability) from Wooldridge’s textbook. However, this book specializes in teaching you econometric methods and not much into causality and econometric techniques for causality. For the causal part of the course, as a *reference textbook*, I recommend *Mastering ‘Metrics: The Path from Cause to Effect* by Angrist and Pischke.

While working with data, we will mostly use the datasets available in the *wooldridge* package in R. These data will correspond to the textbook: the 7th edition of *Introductory Econometrics—A Modern Approach* by Jeffrey Wooldridge. See <https://cran.r-project.org/web/packages/wooldridge/wooldridge.pdf> for documentation on the package and associated data within. You will need to refer to it time and again as you will work through the problem sets and practices that we will do in class.

Finally, even though my primary objective is to teach you econometric methods, to start your journey on the path of becoming an expert of implementing them with real data, there are rarely “textbooks” that achieves everything in the best possible way. To that end, I will follow a combination of various open-source resources listed below that the authors have graciously allowed the academic community to use. Citations of resources I will be drawing from:

- *Statistical Inference via Data Science: A Modern Dive into R and the Tidyverse* (2nd Edition) by Ismai, Kim and Valdivia.
Open-source available at: <https://moderndive.com/v2/>
- *R for Data Science* (2nd edition) by Hadley Wickham, Mine Çetinkaya-Rundel, and Garrett Grolemund.
Open-source available at: <https://r4ds.hadley.nz/>
- *Applied Statistics with R* by David Dalpiaz.
Open-source project: <https://github.com/daviddalpiaz/appliedstats>
- *Introduction to Econometrics with R* by Oswald F, Viers V, Villedieu P, Kennedy G (2020).
Open-source project: <https://scpoecon.github.io/ScPoEconometrics/>
- *Data Science for Economists* by Grant McDermott.
Open-source project: <https://github.com/uo-ec607/lectures>
- *Using R for Introductory Econometrics* (2nd edition) by Florian Heiss.
Open-source available at: <https://www.urfie.net/>

Although this is not necessary, an advanced reading for interested students is *Causal Inference: The Mixtape* by Scott Cunningham. Open-source available at: <https://mixtape.scunning.com/>

More on coding with R

We will use the open-source programming language R to implement the methods we learn in this course. We will write R code in an IDE (*integrated development environment*) called RStudio. Please ensure that both **R** and **RStudio** are installed in your machine following instructions posted in the 1st Canvas announcement.

If you have had no experience with any object-oriented programming language like R before, I recommend that, in the next three weeks, you start familiarizing yourself with the basics of R, even though I will spend the

first week covering the basics. A programming language, like any other language, takes time to get used to. I don't want any student to fall behind just because of this—ultimately, the goal of the course is to build your knowledge and understanding of econometrics. There are several ways to get started:

- Getting started: **Download and install R** and **download and install RStudio IDE**. It is open-source and hence free to use.
 - For students unable to install RStudio, they can use **RStudio Cloud** as an alternative. It offers the same functionality as the desktop version and can be accessed directly at **Posit Cloud**. However, I strongly recommend having it up and running on your own machine, since access to RStudio Cloud will always require a stable internet connection and is only free for 25 hours/month.
- Explore resources in addition to those recommended above.
- Practice regularly: even just an hour a day will help you feel more comfortable with the basics.
- Join study groups: collaborating with peers can make the learning process easier and more enjoyable.

Working with R or any programming language will be difficult initially with a steep learning curve, but it will pay off tremendously in the long run. This is not only because you will learn coding but the ability to implement complicated methods improves analytic and abstract thinking which is a very valuable in the labor market.

To make the most out of the course, I recommend:

- Most important is to ask questions whenever you are in doubt, regardless of what you think about the quality of your question.
 - There is no way you can learn unless you are willing to make mistakes.
 - I want to encourage you to **make mistakes early in class by asking questions**—else the probability of mistakes increases in exams.
- Practicing data analysis in R regularly to build your skills.
- Keeping up with the readings and exercises to reinforce the concepts we cover in class.

Problem sets

Problem sets will be assigned most weeks. Yujie will not accept late problem sets (see more on this under the “Policies” section).

Working format: All assignments will include **both** theoretical calculations, coding exercises requiring you to analyze data sets, and answer other questions based on intuition or on the underlying math of the econometric methods that you learn. To do all of them together in one single file, you will submit your solutions coded and typed up in an *R-notebook* file (.Rmd). Rmd files are designed to allow for code, the output of code and user typed text all in one file. It lets you develop code and record your thoughts by tightly integrating prose and code!

Submission on Canvas: As you will see in the first week, R-notebooks by themselves are hard to grade because the grader will need to download and run it on their own machine. So, after your R-notebook with your code and typed up answers are ready, you will “knit” it to a PDF which you will submit electronically on Canvas. Canvas will not allow you to submit any other file format. To knit to PDF you will first need to install a $\text{\LaTeX}$ distribution for your machine: MacTeX for macOS or MiKTeX for Windows. If knitting to PDF fails, you can knit to HTML and submit. At times installing Java may be necessary depending on your machine. Yujie will discuss R-notebooks and “knitting” them to PDF in the first week in more details.

Expectation: Please make sure you describe all steps you took to get to the answers. You will not receive full credit if you only give the final answer (even if it's correct). You can discuss the problem set with your

classmates, but you need to code up your own file. If you're working with others, you must write their names in the authors section of your R-notebook.

Clarification/doubts on PS: If you have any questions about the problem sets, please reach out to Yujie directly. Yujie will be grading all the problem sets, and I want to ensure consistency in the guidance provided to all students on problem sets. By directing your problem set questions to Yujie instead of me, we can avoid any potential differences in the advice given to students. Again, since Yujie will grade, she will be keeping track of any late assignments. So, if you need an extension due to unavoidable circumstances reach out to Yujie.

Grading

- 5 group problem sets — 10%
 - A group is a group of 2 students, no more. *This is to foster learning from mistakes of one another and hold each other accountable.*
 - * Inform Yujie about your group by the end of the week.
 - One question on each exam will be randomly drawn from any of the PS questions.
- Best of 3 (out of 4) pop-quizzes — 15%
 - Pop-quiz will be taken in the first, or last 15 mins of any class. You could miss the quiz if you come late to the class.
- Class attendance and participation — 10%
 - Attendance sheet will only be available till the first 2 mins of class. Thus arriving 2 mins after the class starts will not count towards your attendance or class participation.
- Discussion session attendance — 5%
 - Attendance sheet will only be available till the first 2 mins of the discussion session. Thus arriving 2 mins after the session starts will not count towards your attendance or class participation.
- Participation will be both voluntary as well as randomly drawn.
- Midterm exam 1 (Thursday, October 8, 2026) — 20%
- Midterm exam 2 (Thursday, November 5, 2026) — 20%
- Final exam (Thursday, December 3, 2026) — 20% (cumulative)
- No exceptions and no re-take of exams allowed.

No collaboration will be allowed on exams.

Overall class grade algorithm:

A	95–100%	C+	78–79.9%
A-	90–94.9%	C	75–77.9%
B+	88–89.9%	C-	70–74.9%
B	85–87.9%	D	60–69.9%
B-	80–84.9%	F	<60%

Engaged Learning Hours

Every one unit/free credit hour course at Clark University has 180 hours of associated engaged academic time. In this course, for the typical student, they may be divided as follows:

Class time	42 hours (14 weeks $\times$ 3 hours)
Discussion session	12 hours (12 classes $\times$ 1 hour)
Reading and class prep	56 hours (14 weeks $\times$ 4 hours)
Problem sets	40 hours (8 $\times$ 5 hours)
Studying for exams	30 hours (3 $\times$ 10 hours)
Total	180 hours

LEEP learning objectives

As part of the Economics curriculum, this course is designed to focus on these learning outcomes: 1) *Knowledge of society*: Economics focuses on the use of quantitative data to understand the economic aspects of society. This course helps you think about how those data are generated and provides you with tools to summarize and synthesize both the description of society and identify potential associations between social phenomena (for example, between education and income). 2) *Intellectual and practical skills*: This course is all about data: learning rigorous methods to distinguish correlation from causation and thinking about what is “real” (the data generating processes) and what is due to chance and then using statistical methods to draw conclusions from it.

Policies

No devices in class:

There is enough empirical evidence that use of laptops and phones in class are distracting to not only the student but also other students. Moreover, most of this class will prepare you to think through econometric problems through mathematical proofs which is best learnt when you write them yourself using pen and paper to improve efficiency of learning. Yujie will walk you through R in the first two weeks in discussion labs where you are required to bring your laptop. However, following that please follow Yujie’s instructions on use of laptops in discussion classes.

AI usage:

There is causal evidence (which we will discuss in class) that shows that “over” usage of AI causes students to perform worse on exams while being stellar on their assignments. Clearly this can and will be linked to worse labor market outcomes. So, even though we cannot monitor your AI usage, I believe that you will minimize your AI usage for your own interest to maximize your labor market outcomes. Given this, with advent of AI the value of critical thinking is going to be probably at its highest. So, what I do not want is that you copy paste your assignment into an LLM and have it produce the solution for you. Submitting answers generated by AI without your own understanding constitutes academic dishonesty.

1. If you completely depend on AI, then you will lose the opportunity to build your skill and knowledge and you will prove to yourself that you are replaceable by AI.
2. In the exam, you will be at a loss unless you have spent enough time thinking about the problems in the assignments by failing to understand the context of the questions.

Students can **efficiently use AI to learn a coding language, but not econometrics**. The word **efficient** is important here. Efficient use of AI is *mostly* using it only in learning to code in R, by asking it to assist you in learning the language, and not to solve your problem sets. This requires you to be proactive and start working on your assignments as early as possible, so that you do not have to give in to the temptation of the easy way out of having the AI solve your assignment thereby worsening your future labor market outcomes.

Late work policy:

Please note that all assignments should be uploaded to Canvas **before** the due date. If you need an extension for an assignment you need to email Yujie (YujiZeng@clarku.edu) **and ask for an extension at least 48 hours before the assignment is due**. Once the 48-hour timeline before the assignment is due is past, **no requests** will be accepted. This will ensure that you can plan to work on the assignments in advance and allow Yujie to post the solutions promptly on Canvas. If you anticipate **substantial** difficulty meeting a deadline, please reach out to me in advance to request the problem set early, allowing you additional time to complete it.

Objective specific emails for efficiency / email policy:

Task	Whom to email
Extension requests on PS	Yujie
Course content	Moshi / Yujie / both
Question on the PS	Yujie
Coding questions	Yujie

Academic integrity:

All students are expected to adhere to Clark's standards of academic integrity; this means that all work must be entirely your own and entirely unique to this course. Plagiarism and other forms of cheating will not be tolerated or excused. For more information, please refer to the university's policy on this issue, available at <https://catalog.clarku.edu/content.php?catoid=32&navoid=2735#academic-integrity> or in the student handbook. If you have any questions about proper citation or other related issues, please don't hesitate to come see me.

Students with disabilities:

Clark University is committed to providing students with documented disabilities equal access to all university programs and facilities. Students are encouraged to register with Student Accessibility Services (SAS) to explore and access accommodations that may support their success in their coursework. SAS is located on the second floor of the Shaich Family Alumni and Student Engagement Center (ASEC). Please contact SAS at accessibilityservices@clarku.edu with questions or to initiate the registration process. For additional information, please visit the SAS website at: <https://www.clarku.edu/offices/student-accessibility-services/>

Title IX:

Clark University and its faculty are committed to creating a safe and open learning environment for all students. Clark University encourages all members of the community to seek support and report incidents of sexual harassment to the Title IX office (titleix@clarku.edu). If you or someone you know has experienced any sexual harassment, including sexual assault, dating or domestic violence, or stalking, help and support is available.

Please be aware that all Clark University faculty and teaching assistants are considered responsible employees, which means that if you tell me about a situation involving the aforementioned offenses, I must share that information with the Title IX Coordinator, Brittany Brickman (titleix@clarku.edu). Although I have to make that notification, you will, for the most part, control how your case will be handled, including whether or not you wish to pursue a formal complaint. Our goal is to make sure you are aware of the range of options available to you and have access to the resources you need.

If you wish to speak to a confidential resource who does not have this reporting responsibility, you can contact Clark's Center for Counseling and Professional Growth (508-793-7678), Clark's Health Center

(508-793-7467), or confidential resource providers on campus: Prof. Stewart (als.confidential@clarku.edu), Prof. Palm Reed (kpr.confidential@clarku.edu), and Prof. Cordova (jvc.confidential@clarku.edu).

FERPA:

The link to Clark's policy regarding student privacy under the Family Education Rights and Privacy Act is available here: <https://www.clarku.edu/offices/security-and-identification-protection/ferpa/>

Disclaimer: The instructor reserves the right to make changes to any information contained in this syllabus at any time during the semester. Changes will be announced, and an updated version of the syllabus will be posted on Canvas and/or distributed to students.